

AUTHENTICATION OF PREMIUM BIOACTIVE-RICH SUGARCANE PRODUCTS

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INTRODUCTION

Sugarcane processing generates by-products such as molasses and vinasse, which can be further valorised into high-value polyphenol-rich sugarcane extracts (PRSEs) for nutraceutical, functional food, cosmetic, and animal feed applications. Australian manufacturers such as The Product Makers (TPM) transform these sugarcane by-products into premium bioactive-rich products. However, their increasing commercial value creates vulnerability to substitution or adulteration with lower-value materials such as sugar syrups or sugar beet molasses, potentially compromising bioactive integrity, provenance, and authenticity. Such adulteration may alter the expected bioactive composition and quality of these products, highlighting the need for reliable methods to verify their authenticity and origin. This study applies SPME–GC–TOF–MS-based untargeted metabolomics to establish volatile chemical fingerprints of sugarcane-derived products and identify potential markers for origin differentiation and authentication.

METHOD

An optimised SPME–GC–TOF–MS workflow was developed and applied to Australian, international, and commercial molasses, vinasse, and PRSEs supplied through TPM. A targeted analyte finding (TAF) method was developed and validated using LECO Target Analyte Find to support consistent detection and data processing of selected volatile compounds. For preliminary geographical comparison, representative Australian and Argentinian raw molasses samples were analysed. Selected volatile compound peak areas were normalised to the internal standard, methyl octanoate, prior to principal component analysis (PCA) and hierarchical clustering analysis (HCA) to assess differences in volatile chemical profiles.

RESULTS AND DISCUSSION

PCA demonstrated clear separation between Australian ($n = 3$) and Argentinian ($n = 3$) raw molasses (**Figure 1**). PC1 and PC2 explained 79.6% and 12.5% of the variance, respectively, accounting for 92.1% of the total variation. HCA further supported this separation, with samples clustering according to geographical origin (**Figure 2**). The heatmap also revealed differences in the relative abundance of several selected volatile compounds between the two groups. These preliminary findings demonstrate distinct origin-related volatile chemical fingerprints.

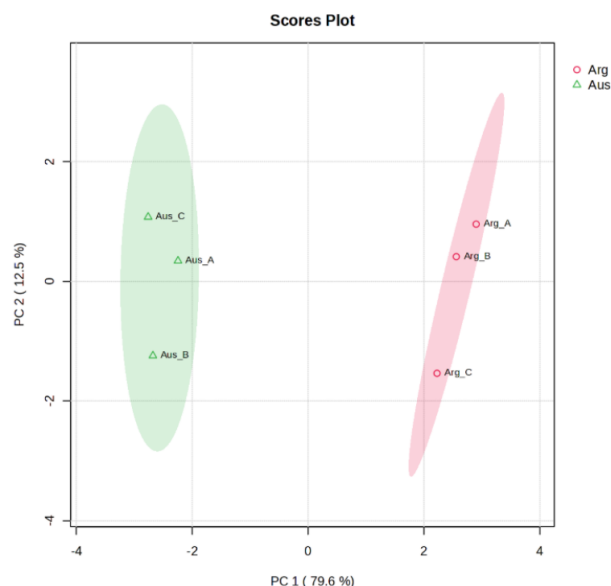


Figure 1. PCA scores plot illustrating separation of representative Australian and Argentinian raw molasses based on normalised volatile compound peak areas.

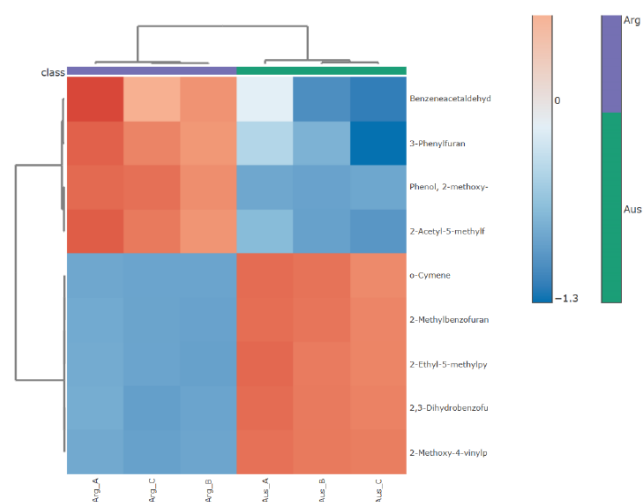


Figure 2. HCA heatmap showing clustering and relative abundance patterns of selected volatile compounds in representative Australian and Argentinian raw molasses.

CONCLUSION

These preliminary findings demonstrate that SPME–GC–TOF–MS can be used as a tool to differentiate sugarcane-derived products based on geographical origin through their volatile chemical profiles. Ongoing analysis of the complete sample set ($n = 84$), comprising Australian, international, commercial, and PRSE samples, will support putative biomarker discovery and the future development of a rapid authentication workflow for industry.